CurrentBatchNumber [SimTalk]

Returns the batch number of the part that was created last if the attribute *GenerateAsBatch* is activated (`true`). Returns the number of created parts if *GenerateAsBatch* is not is activated (`false`).

Type

Read-only attribute

Syntax

```
<Path>.CurrentBatchNumber -> integer
```

Return Value

The return value has the data type *integer*.

Example

```
print MySource.CurrentBatchNumber
```

See also

Generate as Batch [check box]

GenerateAsBatch [SimTalk]

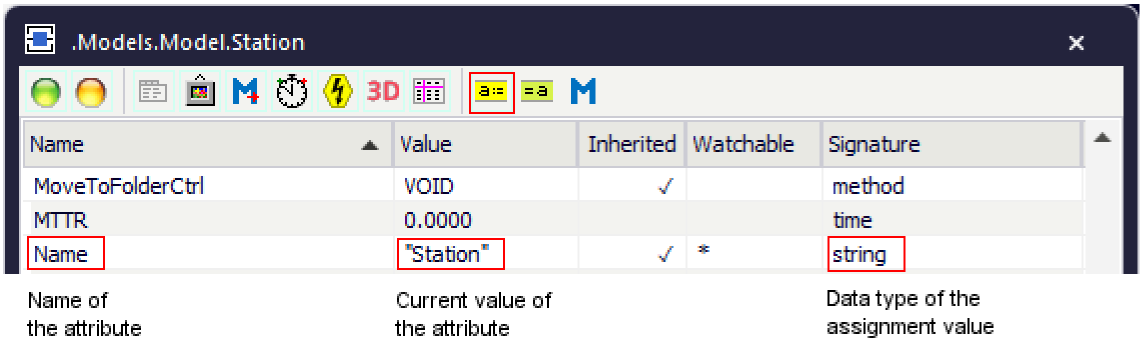
Attributes of the Source

_Attributes of the Source

The *Source* provides:

- The *attributes* listed in the table of contents to the left.
- The **Attributes of All Objects**.
- The **Attributes of the Material Flow Objects**.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.



- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected **Class [general description]**.
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected **Instance [general description]**.

You can set the value of an attribute and you can get its value, either with the check boxes, the text boxes and drop-down lists in the dialog windows or by assigning values to the respective attributes.

- To **set the value** of an attribute, you might, for example, type:

```
MySource.GenerateAsBatch := true
```

- To **get the value** of an attribute, you might, for example, type:

```
print MySource.GenerateAsBatch
posit := MyStation.Cont.XPos
```

Blocking [SimTalk]

Sets if the *Source* designated by <Path> saves the time at which it was supposed to produce the next *MU* (`true`) or not (`false`).

Remarks

When you set *Blocking* to `true`, the *Source* produces the following *MUs* at the next possible point in time, i.e., when the *Source* moved the *MU* that was blocked by the successor, on (`true`).

When you set *Blocking* to `false`, the *Source* creates another *MU* only at the **Time of Creation** which you entered.

Type

Attribute

Syntax

```
<Path>.Blocking:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MySource.Blocking := true
```

See also

Operating Mode [check box]

Time of Creation [Source]

CreationTableActive [SimTalk]

Sets if the *Source* designated by <Path> writes all events, for which it produced *MUs* during a simulation run, into a table (`true`) or not (`false`).

Remarks

The *Source* records the creation events in addition to the resource statistics it collects in any case.

Type

Attribute

Syntax

```
<Path>.CreationTableActive:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MySource.CreationTableActive := true
```

SimTalk

creationTable [SimTalk] - Source

See also

Creation Table [Source]

GenerateAsBatch [SimTalk]

Makes the *Source* designated by <Path> produce *MUs* in a single batch all at once (`true`) or individually one by one (`false`).

Remarks

If you activate **Generate as Batch**, the *Source* produces the entire set of parts all at once at the given start time and attempts to move all of the parts on to the next object in a single lot.

> **Note**
> This setting applies for the **MU Selection** > **Sequence Cyclical**, **Sequence**, **Random**, and **Percentage**.

> **Note**
> If you set **MU Selection** to **Number Adjustable** and if you activate the check box **Generate as Batch**, the **Amount** does not designate the number of *parts*, but the number of *batches* that the *Source* produces.

Type

Attribute

Syntax

```
<Path>.GenerateAsBatch:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MySource.GenerateAsBatch := true
```

SimTalk

CurrentBatchNumber [SimTalk]

See also

Generate as Batch [check box]

MU selection [drop-down list] > **Sequence Cyclical [MU selection]**

Sequence [MU selection]

Random [MU selection]

Percentage [MU selection]

Number Adjustable [time of creation]

Interval [SimTalk] - Source

Sets the *interval* between the times at which the *Source* designated by <Path> produces *MUs*.

Remarks

The behavior of the attribute *Interval* depends on the setting **TimeOfGeneration**:

- For **Time of Creation** > **Interval Adjustable** this applies:

  The attribute *Interval* sets the time that elapses between the creation events of the *Source* designated by <Path>. Specify a distribution and enter the values, which that distribution requires. You can also specify **Const** and enter a constant time.

  The attribute *Start* sets the simulation time at which the *Source* starts producing *MUs*.

  The attribute *Stop* sets the simulation time at which the *Source* stops producing *MUs*. A *Stop* time of `0` (zero) indicates that the *Source* will keep producing *MUs* infinitely.

- For **Time of Creation**> **Number Adjustable** this applies:

  You can specify one of the *Probability Distributions* and specify the values that the selected distribution requires. The distribution determines the point in time at which the *Source* creates the *MUs*.

  The *Source* produces the number of *MUs* at the different times which the random number generator generated at the beginning of the simulation. The determined times are contained within the bounds if you enter a lower bound and an upper bound.

  Specify a **Constant** time to make the *Source* create the number of *MUs* all at one point in time.

Type

Attribute

Syntax

```
<Path>.Start:time
<Path>.Stop:time
<Path>.Interval:time
```

Assignment Value

You can assign a value of data type *time*.

Example

```
MySource.TimeOfCreation := "Interval adjustable"
MySource.Start := 3600
```

```
MySource.Stop := str_to_time("1:00:00:00.00")
MySource.Interval.setTypeAndAttr("Normal",1,500,100,200,900)
```

SimTalk

Start [SimTalk] - Source

Stop [SimTalk] - Source

See also

TimeOfGeneration [SimTalk]

Start [Source]

Interval [Source]

Stop

Interval Adjustable [time of creation]

Interval Adjustable [time of creation]

Probability Distributions

MUSelection [SimTalk]

Sets how the *Source* designated by <Path> selects the *MUs* to be produced.

Remarks

Depending on the setting that you enter, the attribute *Path* either sets the path to the delivery list, the sequence table, the frequency table, or the *MU* class.

To use a copy of the *DataTable* instead of a reference to the *DataTable*, assign the name `.unshare` to the subtable.

Type

Attribute

Syntax

```
<Path>.MUSelection:string
```

Assignment Value

You can assign a value of data type *string*.

You can specify **"Constant"**, **"Sequence Cyclical"**, **"Sequence"**, **"Random"**, **"Percentage"**, or **"Order Controlled"**.

Example

```
MySource.MUSelection := "Sequence"
```

SimTalk

Path [SimTalk] - Source

See also

Store as **Supermarket [check box]**, **Configuration [button]**

MU selection [drop-down list]

Number [SimTalk]

Sets the number of *MUs* which the *Source* designated by <Path> produces.

Remarks

This setting only applies for the **Time of Creation > Number Adjustable** and **Interval Adjustable**.

If you do not enter a **Stop** time for the setting **Interval Adjustable**, *Plant Simulation* produces the amount of parts you entered at the most. The default value of -1 designates an unlimited number of parts to be produced. Otherwise the *Source* produces parts until the **Stop** time is reached and ignores the **Amount** of parts you entered.

For the setting **Interval Adjustable** the *Source* might possibly create more parts than the **Amount** you entered, if the check box **Generate as Batch** is activated. This is because the batches are always produced in their entirety, not in part only

For the setting **Time of creation > Number Adjustable** the value does not designate the amount of *MUs* but the amount of batches/lots which the *Source* produces when you activated **Generate as Batch**.

Type

Attribute

Syntax

```
<Path>.Number:integer
```

Assignment Value

You can assign a value of data type *integer*.

Example

```
MySource.Number := 5
```

See also

MU selection [drop-down list]

Number Adjustable [time of creation]

Stop [Source]

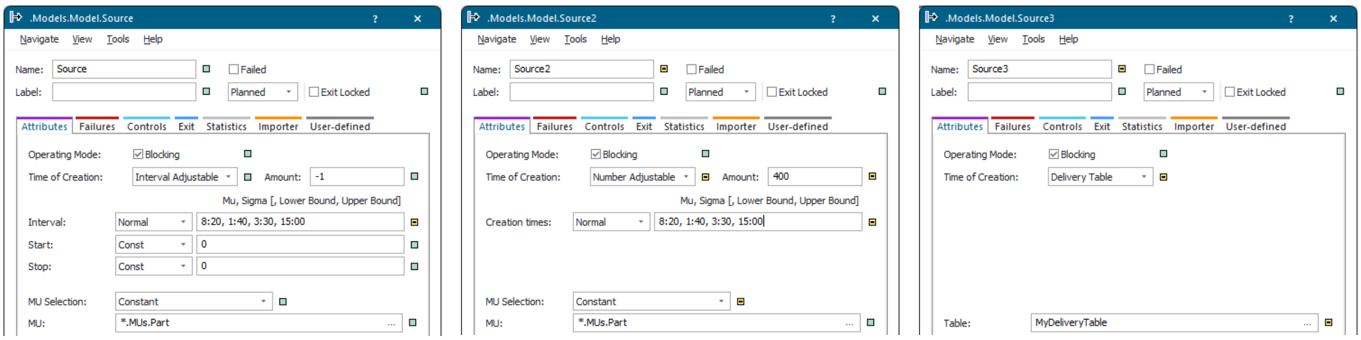
Generate as Batch [check box]

Path [SimTalk] - Source

Sets the path to the delivery list, the sequence table, the frequency table, or the *MU* class of the *Source* designated by <Path>.

Remarks

What the path relates to depends on the settings for **TimeOfGeneration** and **MUSelection**:



> **Note**
>
> The *Source* does not produce parts whose time of creation is in the future when you assign a **Delivery Table** during the simulation.

Type

Attribute

Syntax

```
<Path>.Path:string
```

Assignment Value

You can assign a value of data type *string* or *object*.

Example

```
MySource.Path := "MyDeliveryTable"
```

See also

Time of Creation [Source]/TimeOfGeneration [SimTalk]

MU selection [drop-down list]/MUSelection [SimTalk]

Delivery Table [time of creation]

Start [SimTalk] - Source

Sets the simulation time at which the *Source* designated by <Path> starts producing *MUs*.

Remarks

Start applies when you set the *TimeOfGeneration* to **Interval Adjustable**.

The attribute *Interval* sets the time span between two events at which the *Source* produces *MUs*.

The attribute *Stop* sets the simulation time at which the *Source* stops producing *MUs*. A *Stop* time of zero indicates that the *Source* will keep producing *MUs* infinitely.

Type

Attribute

Syntax

```
<Path>.Start:time
<Path>.Stop:time
<Path>.Interval:time
```

Assignment Value

You can assign a value of data type *time*.

Example

```
MySource.TimeOfGeneration := "Interval adjustable"
MySource.Start := 3600
MySource.Stop := str_to_time("1:00:00:00.00")
MySource.Interval.setTypeAndAttr("Normal",500,100,200,900)
```

SimTalk

TimeOfGeneration [SimTalk]

Stop [SimTalk] - Source

Interval [SimTalk] - Source

See also

Start [Source], text box

Interval [Source], text box

Stop [Source], text box

Interval Adjustable [time of creation]

Stop [SimTalk] - Source

Sets the simulation time at which the *Source* designated by <Path> stops producing *MUs*.

Remarks

A *Stop* time of zero indicates that the *Source* will keep producing *MUs* infinitely.

The attribute *Start* sets the simulation time at which the *Source* starts producing *MUs*, when you set the *TimeOfGeneration* to **Interval Adjustable**.

The attribute *Interval* sets the time span between two events at which the *Source* produces *MUs*.

Type

Attribute

Syntax

```
<Path>.Start:time
<Path>.Stop:time
<Path>.Interval:time
```

Assignment Value

You can assign a value of data type *time*.

Example

```
MySource.TimeOfGeneration := "Interval adjustable"
MySource.Start := 3600
MySource.Stop := str_to_time("1:00:00:00.00")
MySource.Interval.setTypeAndAttr("Normal",500,100,200,900)
```

SimTalk

Start [SimTalk] - Source

Interval [SimTalk] - Source

TimeOfGeneration [SimTalk]

See also

Start [Source]

Interval [Source]

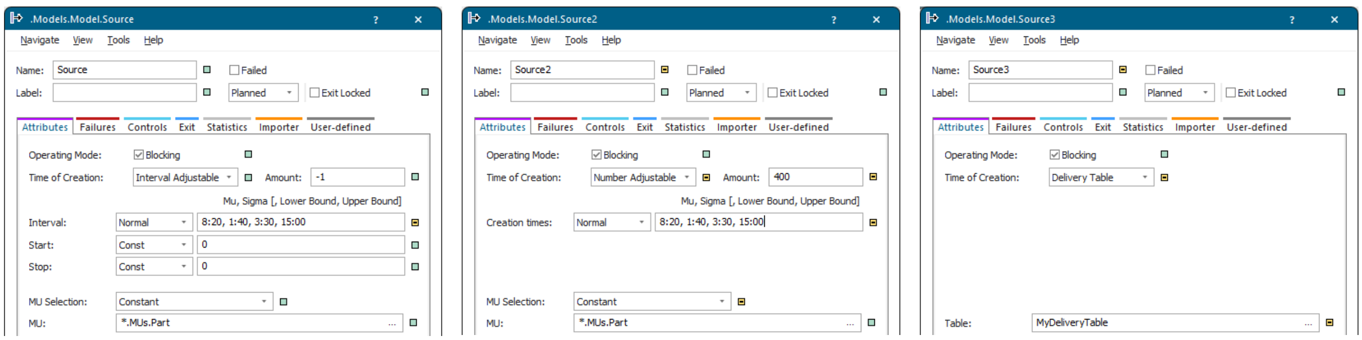
Stop [Source]

TimeOfGeneration [SimTalk]

Sets how the *Source* designated by <Path> produces new *MUs*.

Remarks

Depending on the setting that you enter, the attribute *Path* either sets the path to the delivery list, the sequence table, the frequency table, or the *MU* class.



> **Note**
> The *Source* does not produce parts whose time of creation is in the future when you assign a **Delivery Table** during the simulation.

> **Note**
> *Plant Simulation* processes the *Delivery Table* row by row. For this reason the time should increase from row to row. If *Plant Simulation* processes a row in which the time is located in the past, the part will be created immediately, which is not what you would expect.

Type

Attribute

Syntax

```
<Path>.TimeOfGeneration:string
```

Assignment Value

You can assign a value of data type *string*.

You can specify "**Interval Adjustable**", "**Number Adjustable**", "**Delivery Table**", or "**Trigger**".

Examples

```
MySource.TimeOfGeneration := "number adjustable"
```

```
MySource.TimeOfGeneration := "delivery table"
MySource.path := MyDeliveryTable
```

SimTalk

Path [SimTalk] - Source

See also

Time of Creation [Source]

Trigger [SimTalk]

Sets the internal *Trigger* list of the *Source* designated by <Path> or returns it.

Type

Attribute

Syntax

```
<Path>.Trigger:array
```

Assignment Value

You can assign a value of data type *array*.

Examples

```
var assignedTriggers: object[2]
for var j := 1 to 2
   assignedTriggers[j] := to_str("Trigger",j)
next
   Source.Trigger := assignedTriggers
// sets the trigger list
```

```
print MySource.Trigger // gets the trigger list
// [*.Models.Model.Trigger, *.Models.Model.Trigger2]
```

See also

Trigger [time of creation]

Drain

Drain [object]

Use the object *Drain* for removing the parts from the plant after they have been processed. It usually represents the shipping department of your plant.

Image